Tower Maintenance Flow - From Start to Actual Task Build

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Source folder: /Users/ohadformanair/PycharmProjects/Tower_work/maintenance

Total maintenance tasks loaded	**76**
Non-draw-only tasks (planning cadence)	**63**
Purpose	**Operational flow until building actual executable tasks**

Flow Steps

1) Fill inventory baseline: stock, mounted qty, storage location, serial, min level, and item type.

2) Correlate inventory with app data: component mapping, tools vs consumables, and mounted vs stock.

3) Build maintenance tasks in Builder: task groups, timing triggers (hours/draw/calendar), BOM parts + tools, and work package.

4) Readiness check: pick tasks with available parts first; for missing parts create orders from the same flow.

5) Build actual executable tasks/day packs: preparation, safety, manual pages, and procedure are ready before execution.

6) Execute and record: start task, mark done/wait-for-part, update inventory, history, and reporting.

App Maintenance Workflow (Implemented)

- Unified Maintenance Flow in app: Builder -> Prepare Day Pack -> Schedule + Forecast -> Execute + Records.

- Inventory Center supports stock/mounted model, locations, tools classification, and low-stock ordering.

- Work Package contains Preparation, Safety protocol, Procedure, stop plan, and completion criteria.

- Manual context links task pages and BOM pages for technician guidance during execution.

- Execute lane supports done/wait-for-part actions, readiness checks, and reservation/order bridging.

- Diagnostics + path governance ensure stable paths/config for app deployment and debugging.

Quality Gate Before Execution

- Every task has Task_ID, Group(s), Trigger mode, and interval.
- Every task has BOM split into parts + tools.
- Safety protocol is present and high-fall-risk tasks force NO ENTRY.
- Missing parts generate order candidates before task execution.

- Manual context opens task procedure pages and relevant BOM pages.

Table: Tasks Not Only Per-Draw / Startup (Regular Cadence)

This table excludes draw-only startup tasks so planning can focus on weekly/monthly/3M/6M/hours/calendar/on-condition work.

#	Component	Task	Task ID
1	63mm Furnace System	Cyclic purge after atmosphere exposure (minimum 5 cycles; end in vacuum)	FUR-MNT-012
2	63mm Furnace System	Every 3 months OR 1500 operating hours: check pyrometer alignment & window cleanliness	FUR-MNT-007
3	63mm Furnace System	Every 3 months OR 1500 operating hours: inspect graphite insulation/radiation shield; replace if needed	FUR-MNT-008
4	63mm Furnace System	Every 6 months OR 3000 operating hours: check busbar overlap joints tightness	FUR-MNT-009
5	63mm Furnace System	Every 6 months OR 3000 operating hours: inspect vacuum pump filter cleanliness	FUR-MNT-011
6	63mm Furnace System	Every 6 months OR 3000 operating hours: tower alignment check using plumb line	FUR-MNT-010
7	63mm Furnace System	Monthly: inspect bottom doors for distortion/damage; replace if needed	FUR-MNT-006
8	63mm Furnace System	On-condition: inspect/replace bottom sleeve top cap when eroded (e.g., length reduction >3 mm)	FUR-MNT-015
9	63mm Furnace System	On-condition: replace heating element when voltage trend increases (e.g., >2 V above new)	FUR-MNT-014
10	63mm Furnace System	Weekly: test water and gas flow interlocks	FUR-MNT-005
11	Bare Fibre Diameter Gauge	Align bare fibre diameter gauge to fibre line using tightened line	ALGN-003
12	Bottom Centering Pulley + Break Detector	Routine: ensure X/Y slides run free and true	BP-MNT-001
13	Bottom Centering Pulley + Break Detector	Weekly: check pulley rotates freely, bearing play, and clean V-groove	BP-MNT-002
14	Cane / Rod Puller (Caterpillar)	Periodic: check carriage movement, belt tension, and lubricate drive linkages	CP-MNT-002
15	Cane Puller	Align cane puller belts to fibre line; record micrometer baseline and offset for cane diameter	ALGN-006
16	Capstan	Set capstan offset (nominal 50 mm) and mark/bolt capstan position	ALGN-002
17	Capstan System	As needed: adjust urethane belt tension (cam plate)	CAP-MNT-003
18	Capstan System	Every 3 months: check bearings are not worn (sealed for life)	CAP-MNT-002
19	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	As needed: replace HEPA filter element (front face access; tighten nuts diagonally)	CAS-MNT-006
20	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	As needed: replace pre-filter element (with seal inspection and correct airflow direction)	CAS-MNT-005
21	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	Every 6 months: replace pre-filter; re-adjust speed control; replace HEPA if airflow remains incorrect	CAS-MNT-004
22	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	Monthly: inspect wear/damage, verify intakes clear, verify airflow uniform & at optimum; balance dampers	CAS-MNT-003
23	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	When switching on: confirm fibre position in bare fibre gauge window stays centered; correct if needed	CAS-MNT-002
24	Coated Fibre Diameter Gauge	Align coated fibre diameter gauge to fibre line using tightened line	ALGN-010

#	Component	Task	Task ID
25	Coating System (Resin Delivery Block)	Align coating resin delivery block to fibre line; record X/Y micrometer readings	ALGN-007
26	Concentricity Monitor	Align concentricity monitor optics and laser aim to fibre line; fine-tune during draw	ALGN-008
27	Draw Tower – Alignment (overall)	Initial setup: establish fibre line using plumb line through furnace to datum pulley	ALGN-001
28	Draw Tower – Alignment (overall)	Periodic verification: check fibre line alignment with tightened line through key stations	ALGN-011
29	Dual Drum Winder	3■monthly: grease drum bearing housings	LUB-DDW-002
30	Dual Drum Winder	Monthly: lubricate clamps, guide pulleys, layering arm pulleys	LUB-DDW-001
31	General Tower Components	Daily: cleanliness, smooth operation, pulley condition	LUB-GEN-001
32	General Tower Shafts & Slideways	Monthly: wipe bright shafts and slideways with light oil	LUB-GEN-002
33	Guide Pulley + Fibre Tension Gauge	Monthly (or as required): check tension gauge calibration	TG-MNT-002
34	Guide Pulley + Fibre Tension Gauge	On-condition: return load cell to manufacturer if erratic / cannot calibrate	TG-MNT-003
35	Guide Pulley + Fibre Tension Gauge	Weekly: check pulley rotation, bearing play, and clean V-groove	TG-MNT-001
36	He Cooling System	Align He cooling tube entry/exit orifices to fibre line; balance ram stops	ALGN-004
37	Helium Cooling System	Monthly: lubricate slide bearings on cooling tubes	LUB-HCS-001
38	Linear Dancer	Monthly: lubricate dancer pulley spindles and bearings	LUB-LD-001
39	Mini Capstan	Align mini capstan pinch roller to fibre line (coarse + fine adjustment)	ALGN-005
40	Mini Capstan	Weekly: inspect cleanliness, roller bearings, carriage bearings, glass embedment, lubricate, check pneumatic leaks	MC-MNT-002
41	Mini Capstan	Weekly: lubricate carriage guides and roller bearings	LUB-MCAP-001
42	Preform Feed Assembly	3-monthly: grease ball screw	PF-MNT-005
43	Preform Feed Assembly	3-monthly: grease linear rails	PF-MNT-004
44	Preform Feed Assembly	3-monthly: lubricate X-Y slide bearings	PF-MNT-003
45	Preform Feed Assembly	3-monthly: lubricate ball screw support bearings	PF-MNT-006
46	Preform Feed Assembly	3-monthly: lubricate clamp moving parts	PF-MNT-002
47	Preform Feed Assembly	Daily: inspect cleanliness and smooth operation of clamp and X-Y slides	PF-MNT-001
48	Preform Feed Module	3■monthly: grease linear bearings and ball screw (if required)	LUB-PFM-001
49	Preform Feed Module	6■monthly: lubricate actuator O■ring seals	LUB-PFM-002
50	Pressurised Coating System	Monthly: lubricate applicator gimbals and X■Y adjustment mechanism	LUB-PCS-001
51	Take Up Winder	Every 3 months: grease rotating shaft; wipe & grease guide rails (thin film only)	TUW-MNT-002
52	Take Up Winder	Every 6 months: inspect drive train (belt tension + contamination/wear/damage)	TUW-MNT-003

#	Component	Task	Task ID
53	UV Curing System	After full preform draw: inspect quartz liner tube for discolouration (replace if needed)	UV-MNT-002
54	UV Curing System	Align UV lamp entry/exit irises to fibre line (hexapod turnbuckles)	ALGN-009
55	UV Curing System	Every 500 operating hours: clean lamp module, reflector, bulb and RF screen	UV-MNT-003
56	UV Curing System	Monthly: lubricate UV lamp VAM pivot hinges	LUB-UV-001
57	UV Curing System – PSU	Every 3 months: remove dust from PSU units and replace air filters if required	UV-MNT-004
58	Wet-on-Dry Coating System	Applicator alignment to tower axis check	WOD-MNT-008
59	Wet-on-Dry Coating System	Applicator/delivery block levelness check	WOD-MNT-005
60	Wet-on-Dry Coating System	End-of-draw housekeeping	WOD-MNT-004
61	Wet-on-Dry Coating System	Resin line purge after refill / dip-tube exposure	WOD-MNT-002
62	Wet-on-Dry Coating System	Water circuit checks (weekly) + flush (annual)	WOD-MNT-007
63	Wet-on-Dry Coating System	Weekly: joints leakage check + line purge + cleaning	WOD-MNT-006